



KrishokChat: A Safety-First Agentic Agricultural Advisory System for Bangladeshi Smallholder Farmers

CAPSTONE
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NSU ECE

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85,979

benchmark instances, 4 tracks

1.9×

Bengali QA over best zero-shot

95–97%

on-device crop diagnosis

<30 ms

offline Android inference

2 paisa

marginal cost per AI answer

16123

human escalation, always free

The Problem

Bangladesh's 47 million farming households have no reliable advisory in their own language. Undetected crop disease costs 20–40% of yield. A wrong pesticide dose can poison a family and contaminate the water supply — and a generic chatbot, asked for a dosage, will always answer, right or wrong.

The Idea: An Evidence-Grounded Trust Stack

Not a chatbot with an agriculture prompt. A layered trust stack, built on our own research: official publications → machine-readable knowledge graph → provenance-verified retrieval → a safety gate **before** the model ever runs → dosage verification → an answer with citations, or human escalation.

Every layer exists because our measurements showed it must (column 4).

Impact: Safety, Environment, Access

Health. Unsafe agrochemical queries are refused and escalated — to the 16123 Krishi Call Center and 999 — in every tier of the product. Safety is never paywalled.

Environment. Source-verified dosing cuts over-application of pesticide and fertiliser. Independent Bangladesh sensor-irrigation trials report up to 63% water and 50% urea savings — the direction our soil-vision lane targets.

Access. Runs in Bengali — standard and dialect — with voice input, offline on the mid-range Android phones farmers already own.

What Exists vs What We Added

Capability	Generic	K.B.	Ours
Safety check before retrieval	—	—	●
14-field dosage verifier	—	—	●
Audit log + analytics panel	—	—	●
Bengali + 6 dialects	○	○	●
Published 85,979 benchmark	—	○	●
Photo → grounded treatment	○	—	●
Field soil dataset	—	—	●

● present ○ partial — absent K.B. = KrishokBondhu

Ongoing & Next

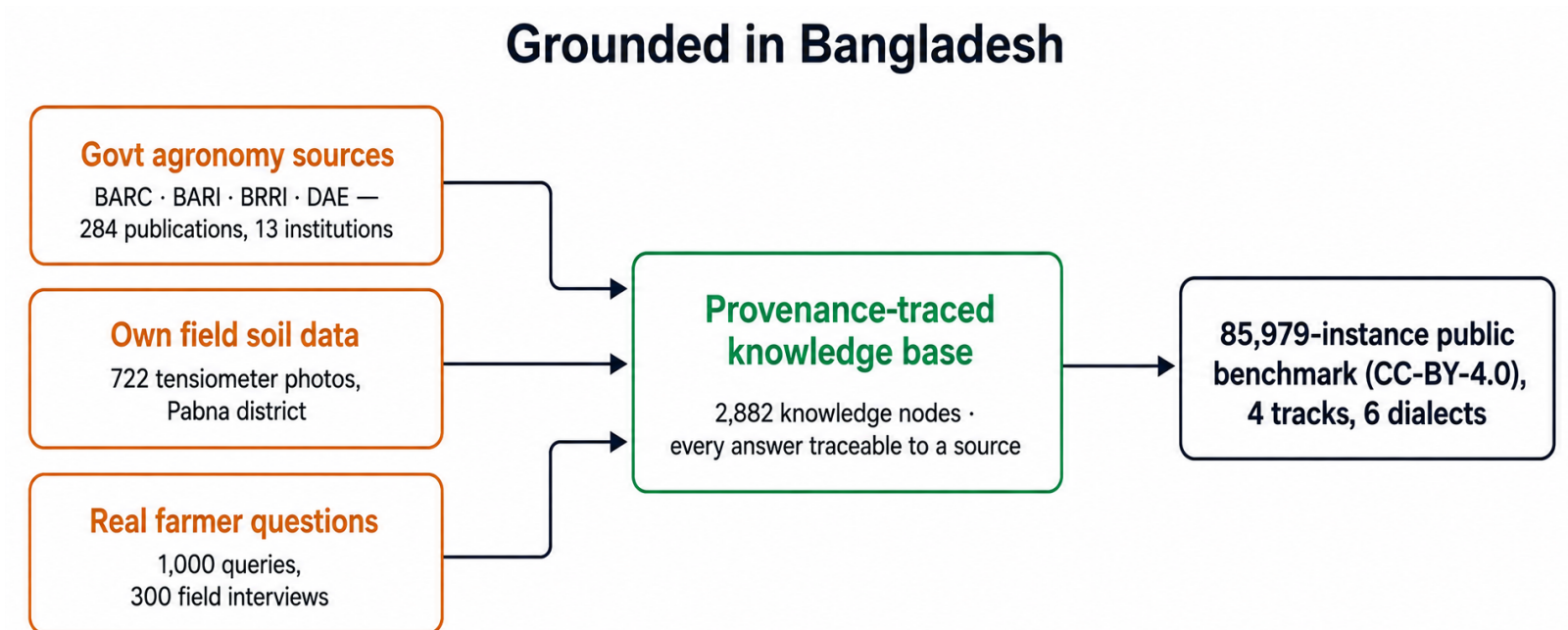
Voice output for low-literacy users · extended dialect coverage · soil-vision refinement toward irrigation advisories · district-level pilot with an extension office.

The Foundation: Four Research Assets

Asset	Scale	Ground truth & provenance
Benchmark, 4 tracks	85,979 instances	General QA 28,993 · Treatment 11,224 (66.3% chemical-bearing) · Table 25,650 · Safety 20,112 — every instance citation-traced
Knowledge graph	2,882 nodes · 17,501 triples	19,768 entities extracted from 284 publications · 13 institutions (BARC, BARI, BRRI, DAE, ...)
Farmer-language benchmark	1,000 queries	300 face-to-face interviews (Rajshahi & Natore) + farmer groups · 6 dialects
Soil-moisture dataset	722 field RGB images	paired tensiometer readings 0.0–21.5 kPa · 6 USDA textures · 14 crops · 46 leakage-checked series · Pabna, May 2026

All released CC-BY-4.0 on Hugging Face.

Grounded in Bangladesh



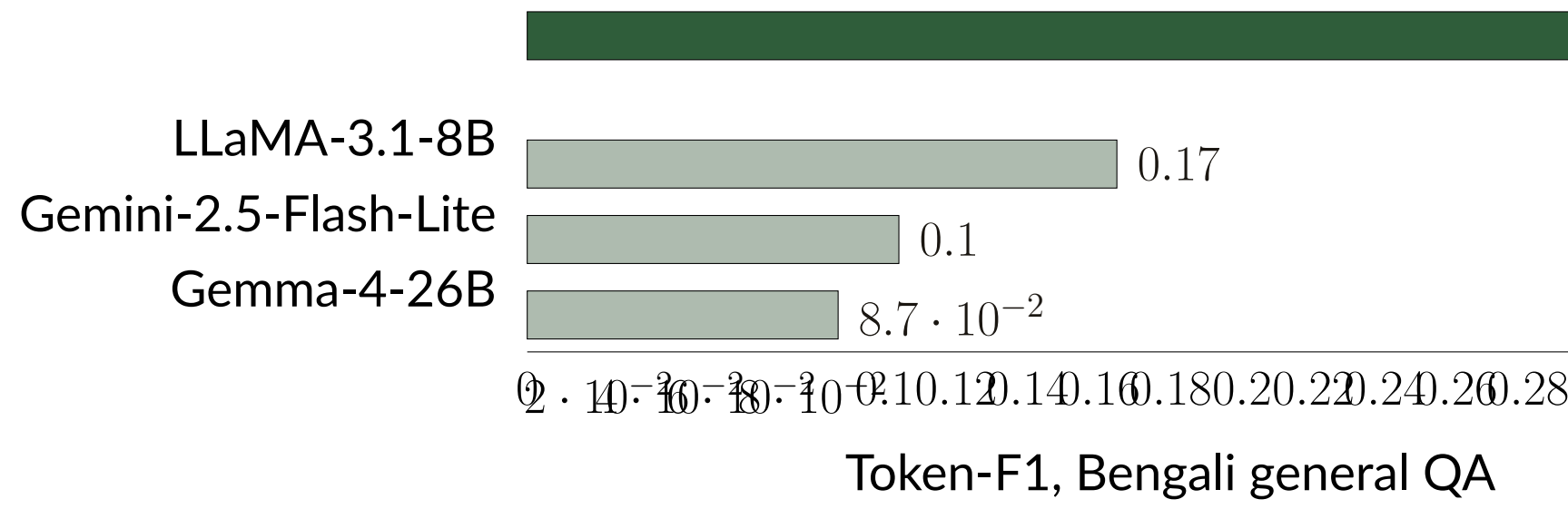
From official publications to provenance-traced retrieval nodes.

Built From the Field, Not the Web



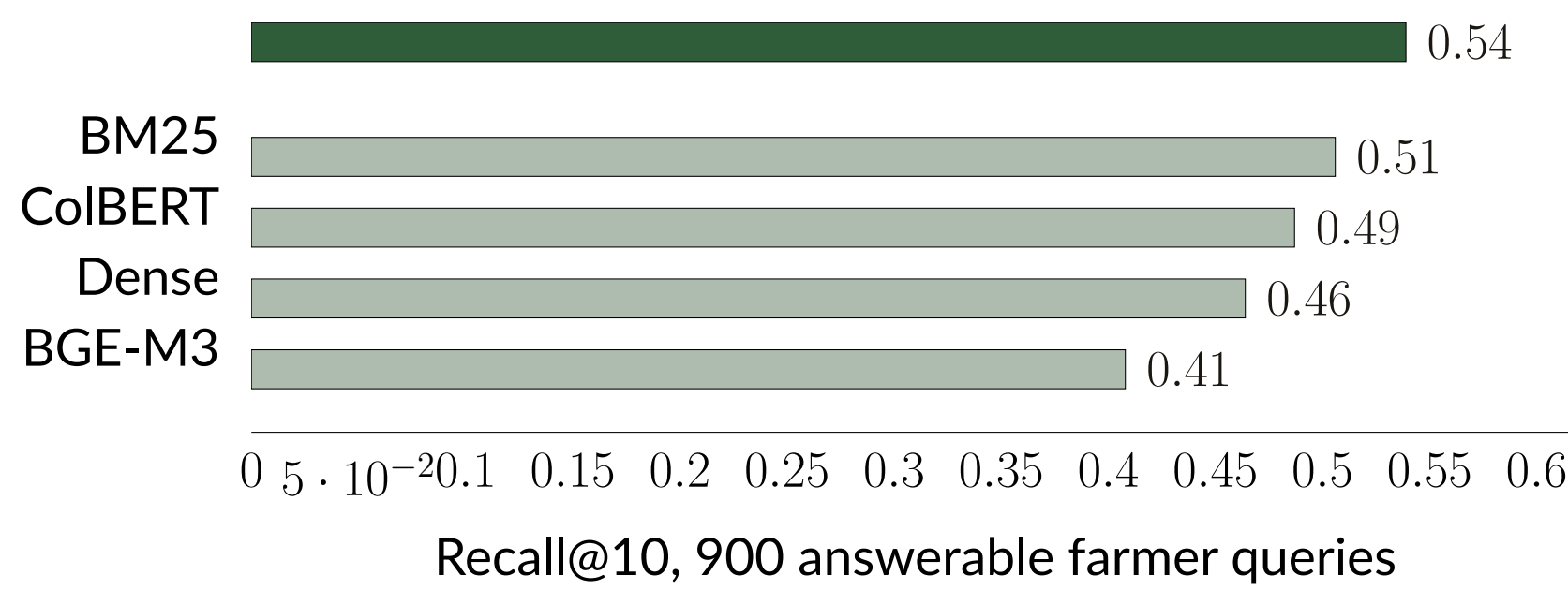
- **Page-level provenance** — every answer traceable to a publication and page.
- **Physical ground truth** — tensiometer kPa readings, not annotator opinion.
- **Series-stratified splits** — no near-duplicate leakage between train and test.

Our On-Device Model Beats Far Larger Ones



1.9× the best zero-shot baseline ($p \approx 6.7 \times 10^{-5}$). LoRA fine-tune of Gemma-4-E4B, 4-bit quantised — deployable on-device.

Retrieval: Hybrid Beats Every Single Method



+9% over dense retrieval ($p < 0.001$, Wilcoxon).

Crop Disease, On-Device

Brassica (11 classes)	97.29%
Corn (4 classes)	97.23%
Rice (8 classes)	96.49%
Potato (3 classes)	95.04%
Wheat (live sweep)	88.00%

436/437 grounded Bengali treatment (99.8%)
<30 ms on-device, Android

Safety Behaviour

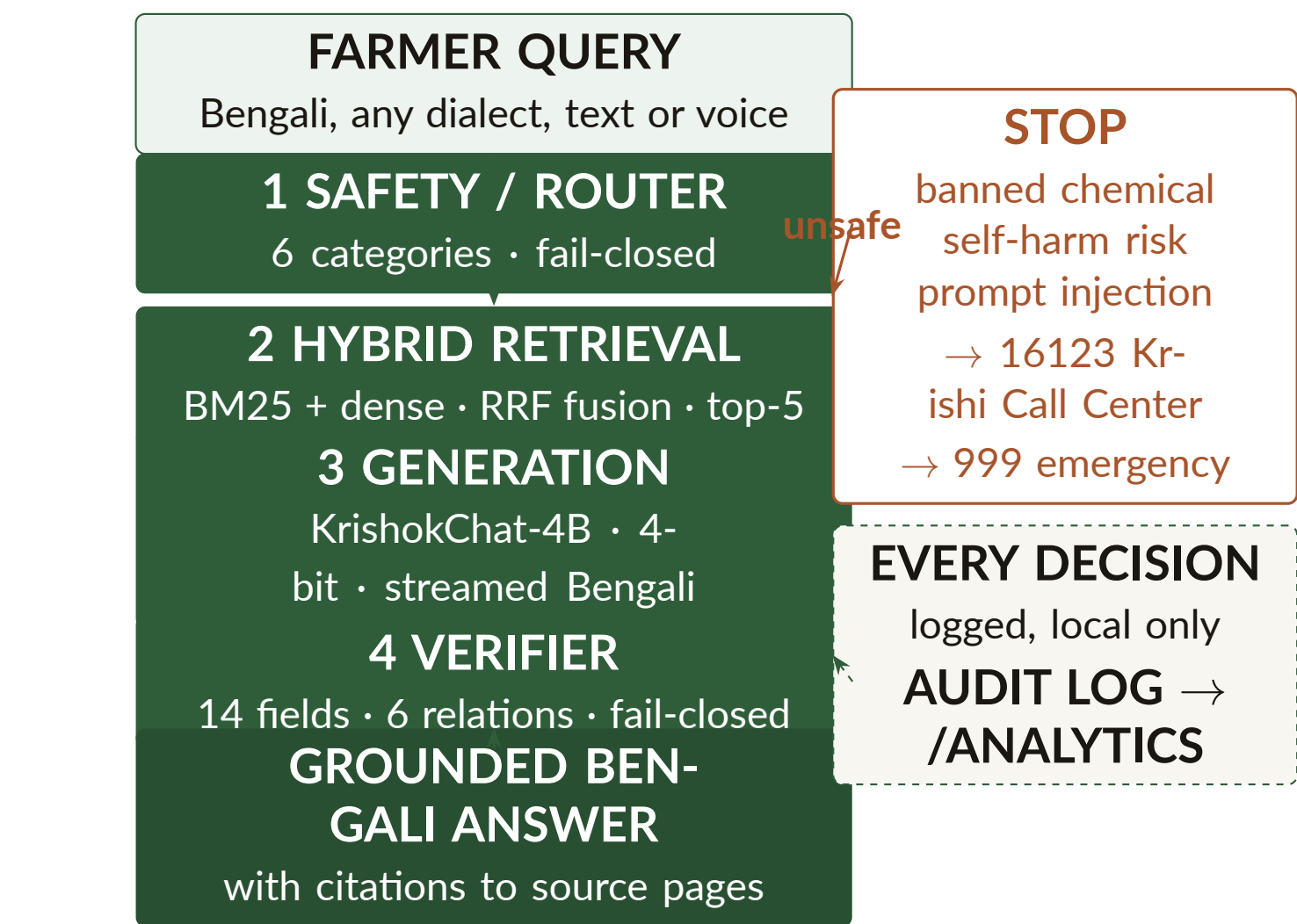
1 of 323 unsafe test queries leaked through — 0.31% non-compliance. Fail-closed by design: missing evidence means abstain.

Test sets: benchmark held-out split · 900 answerable queries · per-crop held-out images.
Disease = classification top-1.

Four Measured Failures → Four Design Decisions

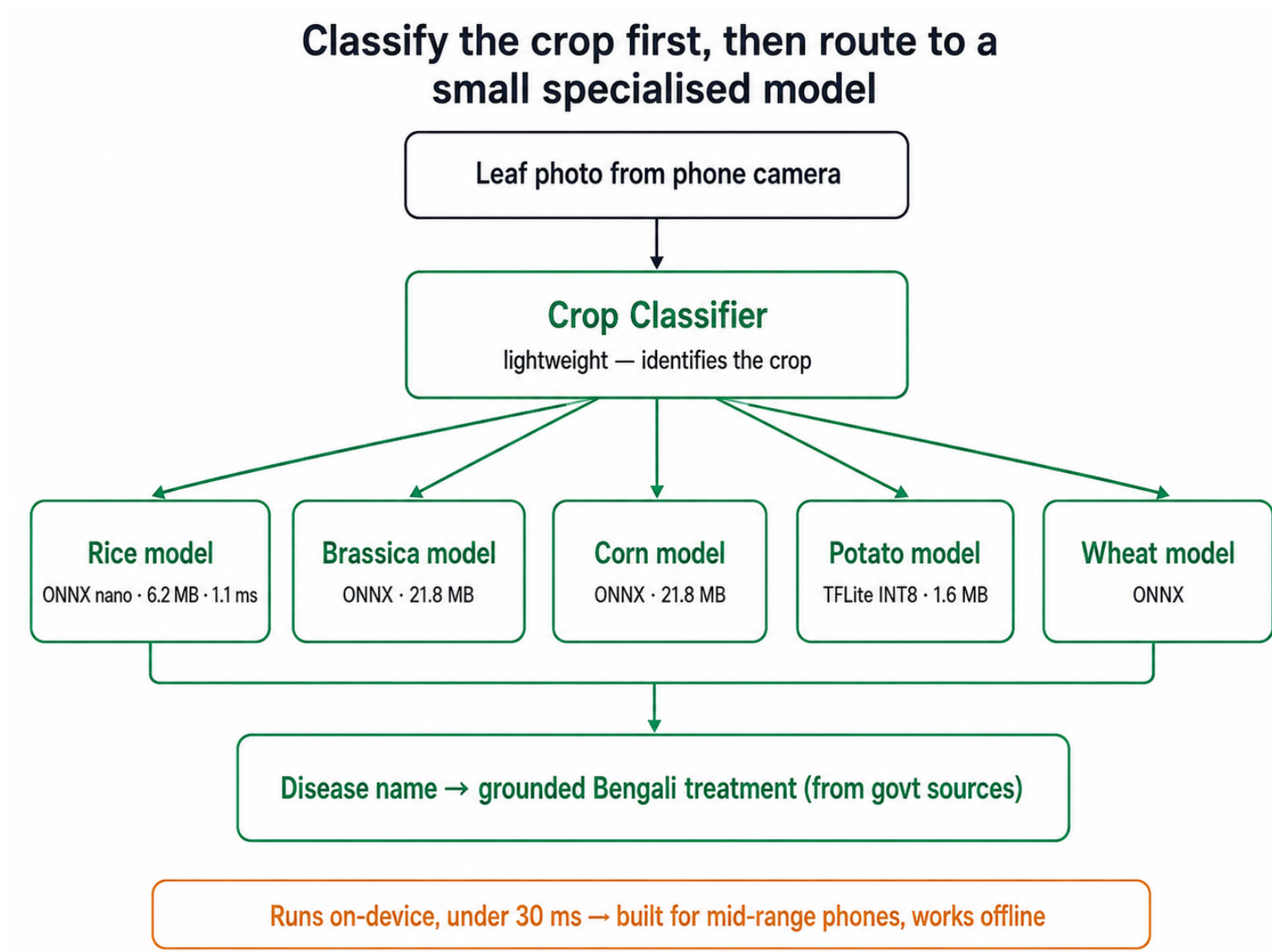
Measured failure	Our design answer
LLMs hallucinate chemical dosages at 4.05–7.00% even with gold evidence	Safety gate before retrieval + 14-field fail-closed verifier
Dense retrieval collapses on farmer language: R@10 0.970 formal → 0.093 colloquial	Hybrid BM25 + dense, RRF fusion
92% of formal entities never appear in farmer queries — পাতা হলুদ হয়ে যাচ্ছে ≠ "Yellow Mosaic Virus"	Register-aware retrieval across 6 dialects
The national helpline answered 92,094 calls last year	AI triage: auto-answer routine, escalate high-risk to humans

System Architecture



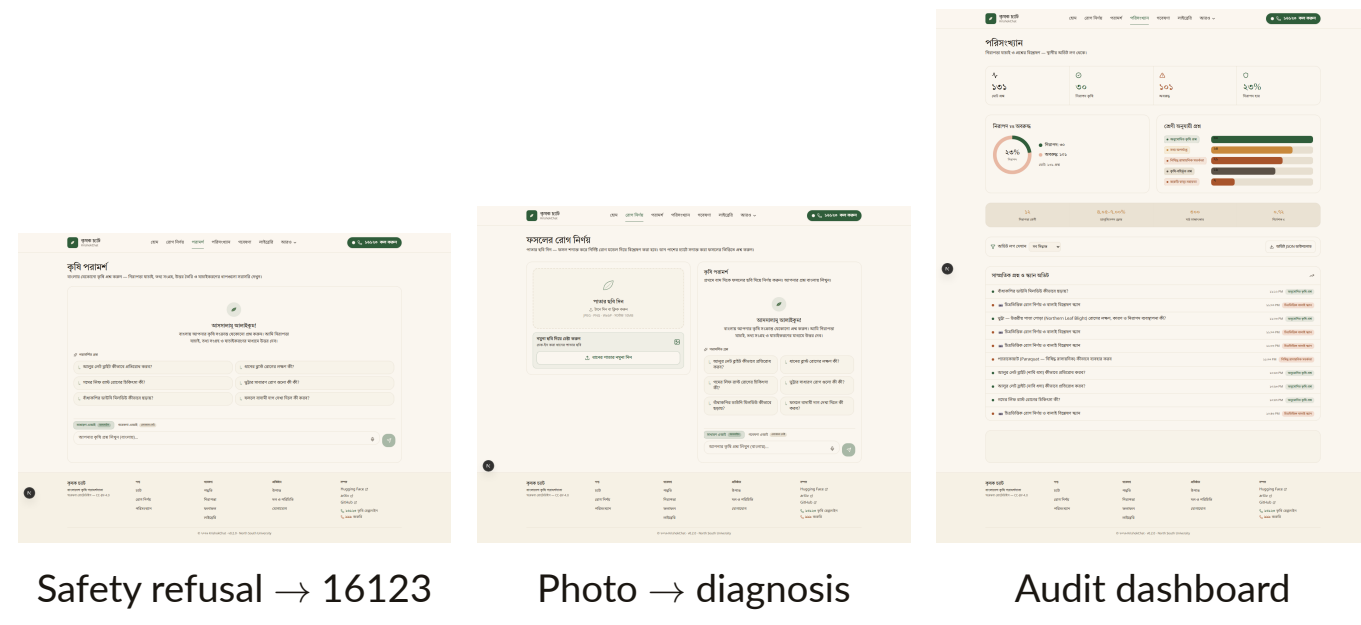
Unsafe categories stop at stage 1 — retrieval, generation and verification never run.

Vision: Classify, Then Route



One photo → crop router → a small per-crop classifier → grounded Bengali treatment. The whole vision stack ships inside the app — no server, no network needed.

The Product --- Krishi Patrak (কৃষি পত্রিকা)



Bengali-first web + Android: live agent trace on every query, voice input, sunlight mode, offline-capable.

Market Readiness

What ships today. Full-stack web app and Android app · on-device disease models that run with no network · live safety pipeline with audit dashboard · 437-entry treatment library verified end-to-end in live testing.

Why offline-first wins this market. 73.4% of households own a smartphone, but only 43.6% of rural residents use the internet. An advisory app that needs the network loses half its users on arrival.

Scaling path. Launch on a hosted API (≈2 paisa per answer); at volume, one rented GPU (≈\$497/month) serves ≈13 million answers monthly.

Business Model: Free for Farmers, Paid by Institutions

FREE vs PREMIUM tier diagram (GPT-generated, 4:3)

FREE: mapped answers from the grounded KB (no LLM — cannot hallucinate) + on-device photo diagnosis + safety gate + offline PREMIUM: everything in Free + open-ended conversation on our fine-tuned model + priority + voice

banner: safety gate and 16123 escalation identical in both tiers

B2G. AI triage front-end for the Krishi Call Center — routine questions answered automatically, high-risk cases escalated with a full audit trail. The audit log is the trust asset for DAE / a2i.

B2B. Analytics dashboard for agro-dealers, seed companies and NGOs: district-level insight into what farmers ask and which diseases surge. (ACI's Fosholi monetises its 2.6M users the same way.)

Licensing. The 85,979-instance benchmark and 722-image soil dataset — CC-BY-4.0 for research, commercial licence for agri-fintech and insurers.

Papers, Code & Data



Dataset (Hugging Face) Code (GitHub)